

Valence is more prominent than truth-conditional meaning

Morgan Moyer[†], Anouch Bourmayan[‡], Isidora Stojanovic[†], Brent Strickland^{*}

[†]Universitat Pompeu Fabra, [‡]Sorbonne Université, ^{*}Institute Jean Nicod, ENS-PSL / CNRS

morgancolleen.moyer@upf.edu,  mcmoyer11, <https://mcmoyer11.github.io/>



Institut | Nicod



What is the relative significance of Valence versus Truth-Conditional info in lexical processing?



Val First

Valence is processed first and is the most prominent feature.

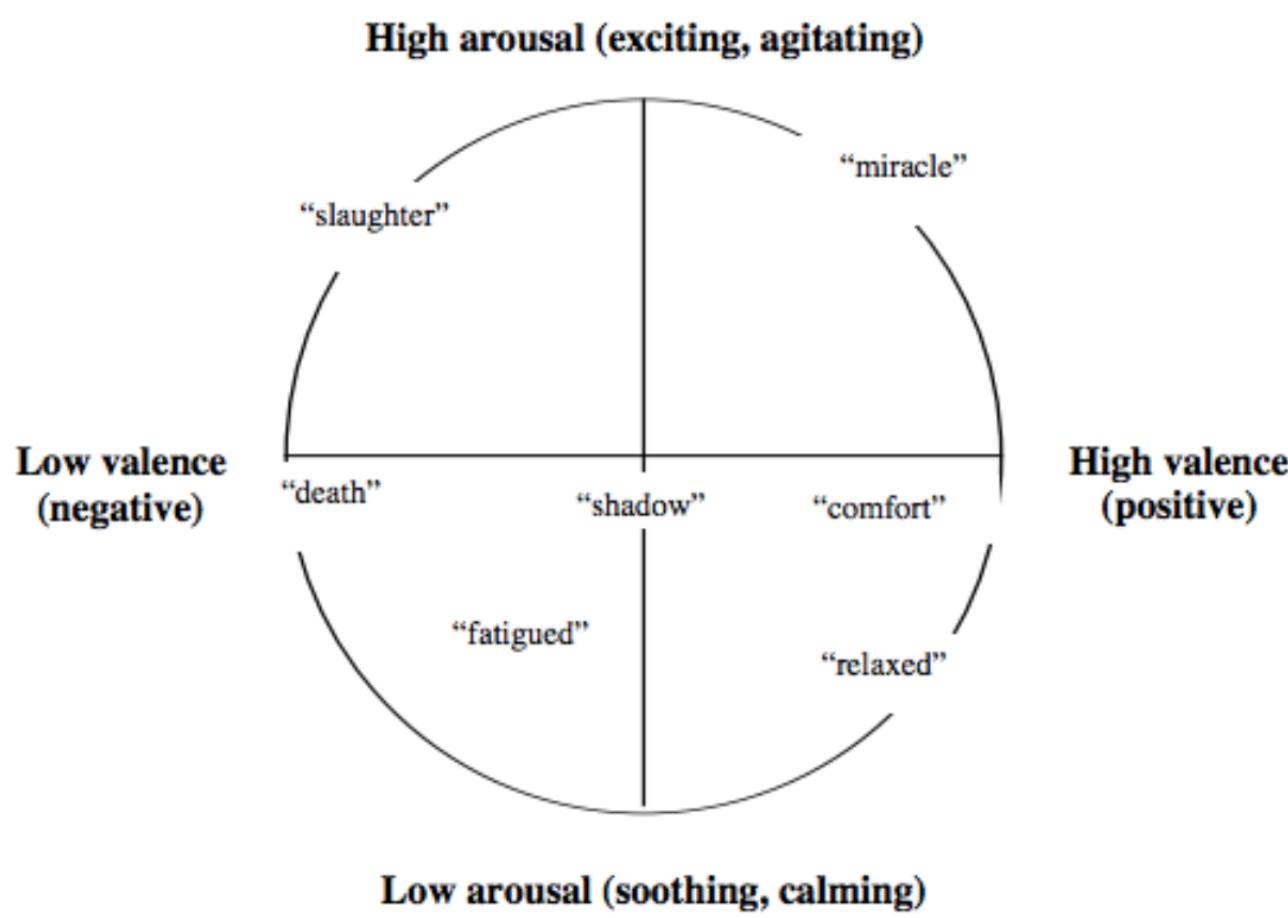
Tru-Con First

Truth-conditional meaning is processed first and is more prominent.

Val Prominence

Valence is usually more prominent than most truth-conditional features, but not necessarily all.

What do Linguists and Psychologists say?



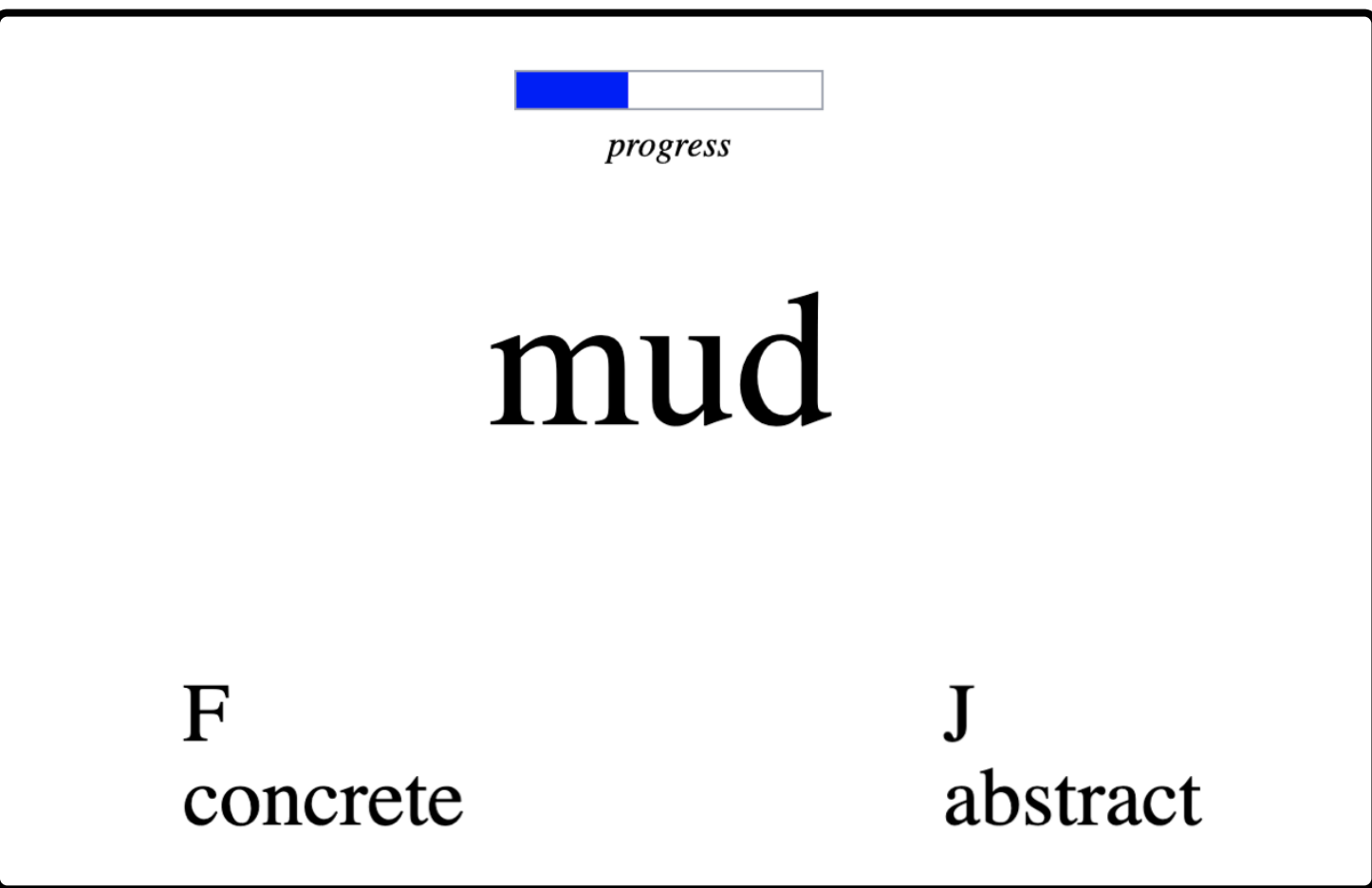
IN SEMANTICS Meaning is truth-conditions [e.g., 3-9]

IN PSYCHOLOGY Concepts located in affective space [10-16]; but disputed empirically, for VFH: [17-26], for TCFH: [27-35]

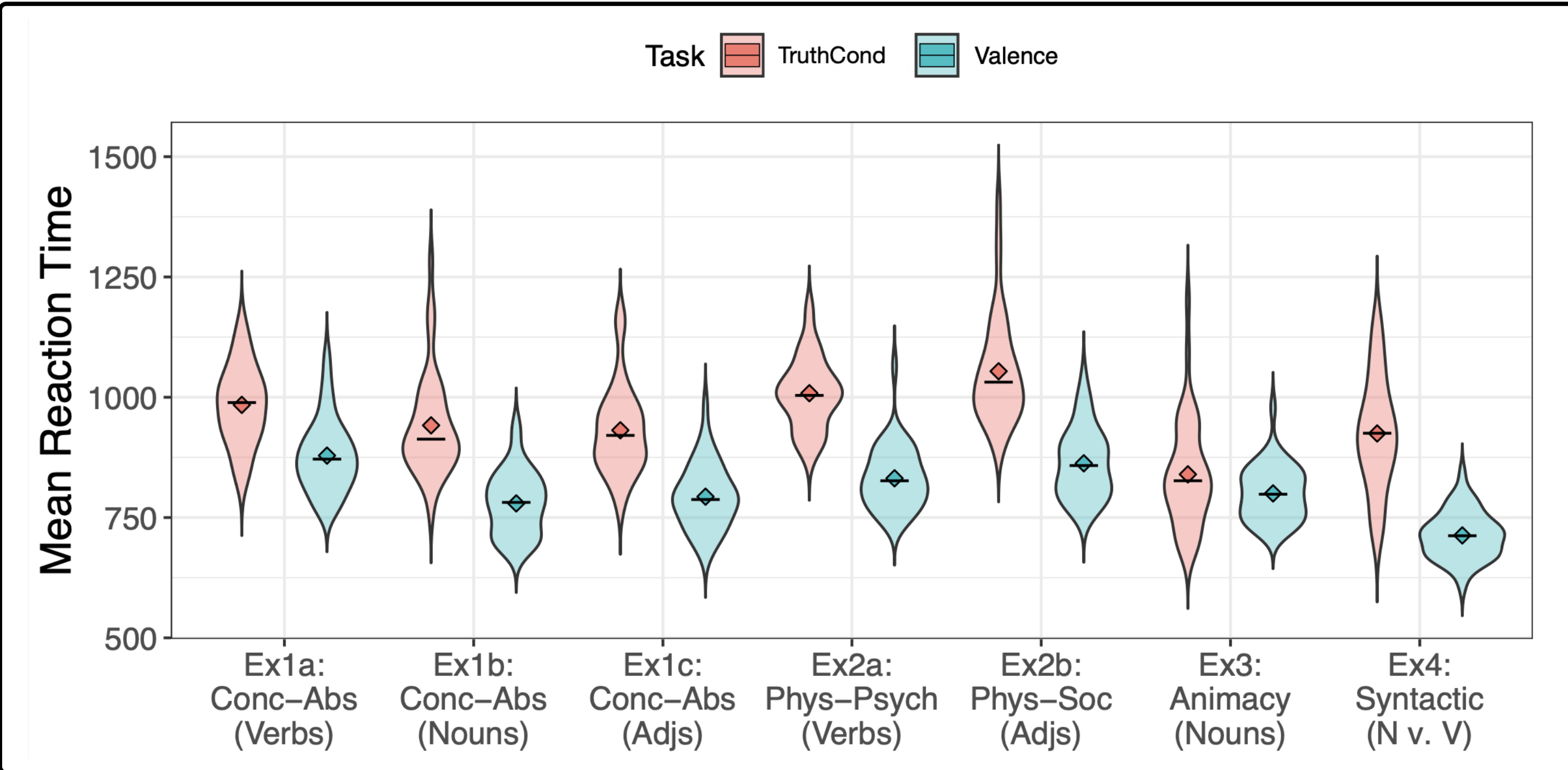
We pit Valence against many Truth-Conditional Foils:

- **Concrete/Abstract** (Exps 1a-1c) [44-54]
- **Physical vs. Psychological/Social** (Exps 2a-2b) [65-70]
- **Animacy** (Exp 3) [71-78]
- **Syntax** (Exp 4) [e.g., 79]

Exps 1-4: Speeded Categorization (n=270)



- **Task:** 2x1 within-subjs design: **Valence**, **TruthCond**
- **DM:** forced choice
- Words normed for feature values [eg. 15,82-83]

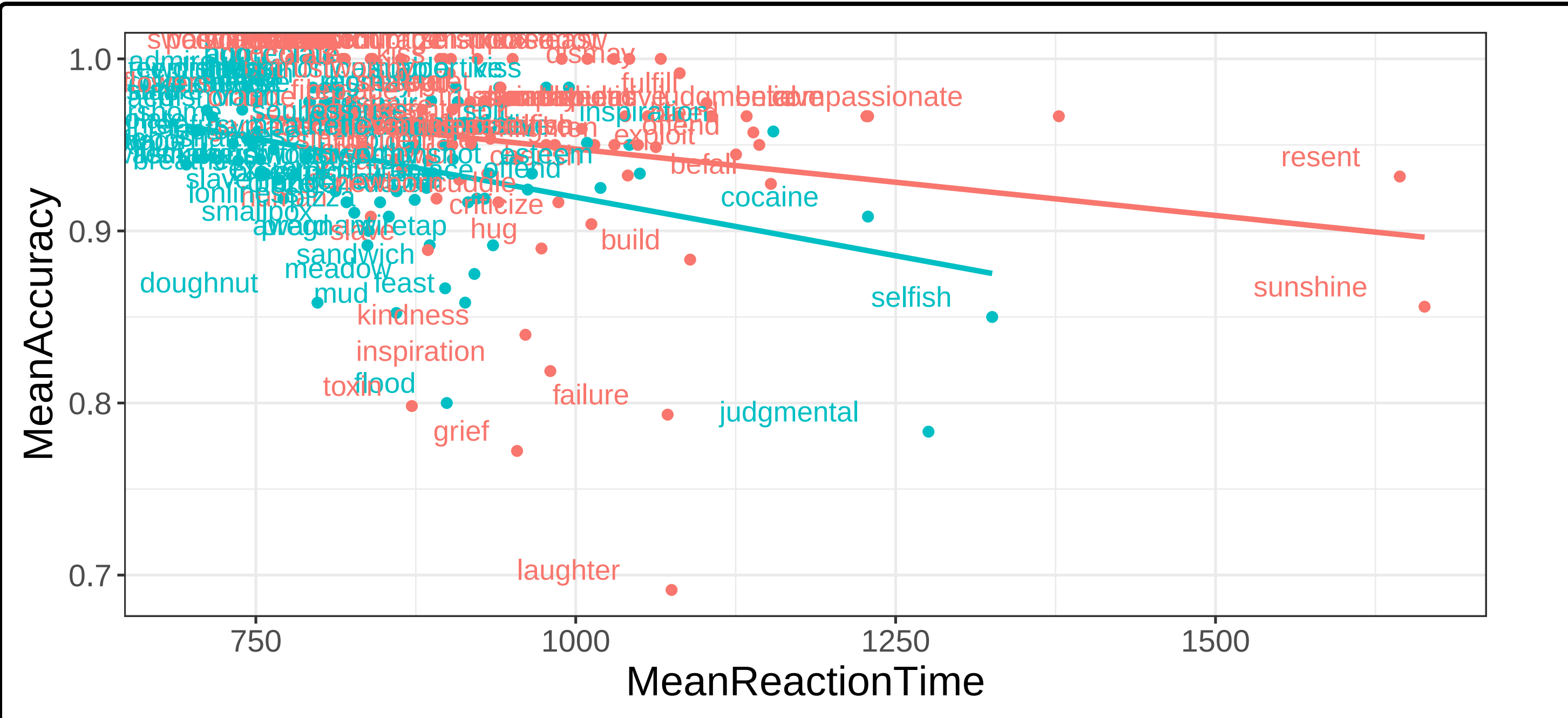


Speeded Categorization Results

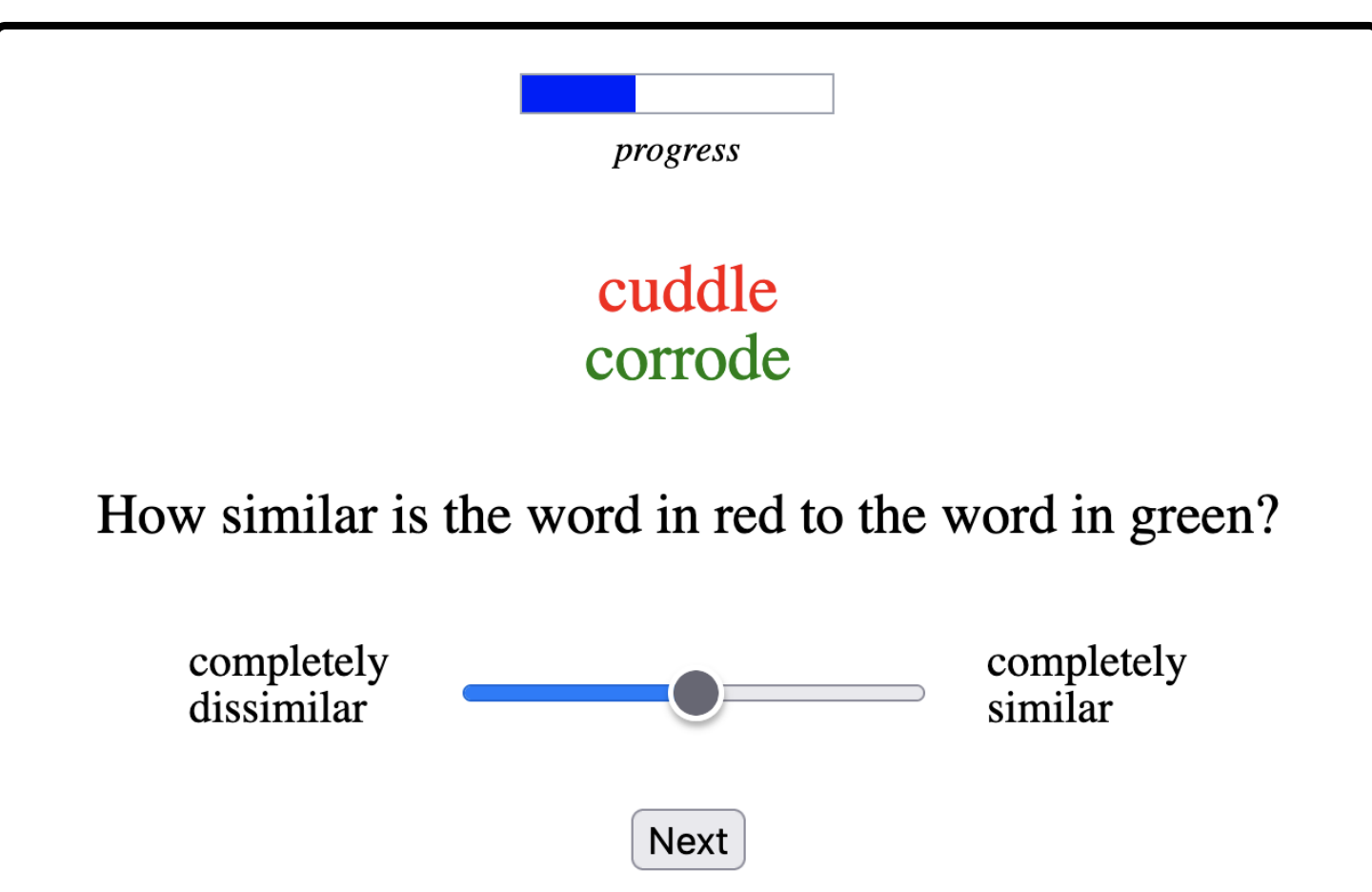
Valence < **TruthCond** except **Animacy** (Ex3)

$\text{lmer}(\text{LogRT} \sim \text{Task} + (1 + \text{Task} | \text{ID}) + (1 + \text{Task} | \text{Word}))$

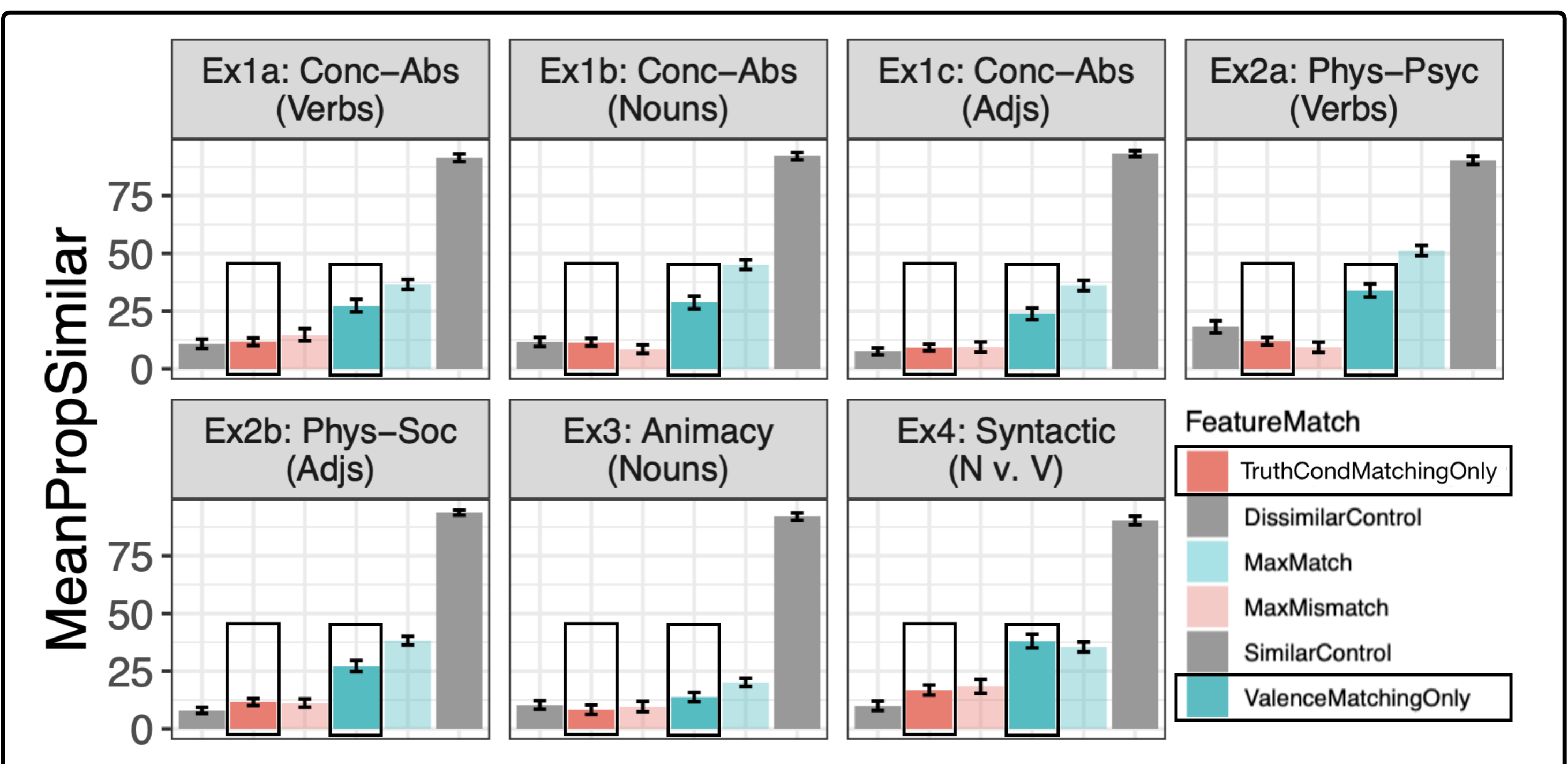
No strong RT~Accuracy relation in high agreement words:



Exps 5-8: Similarity Judgments (n=350)



- **FeatureMatch:** 4 x 1: **ValMatch**, **TCMatch**, **MaxMismatch**, **MaxMatch**
- **DM:** slider from 1-100
- Paired words from Exps1-4

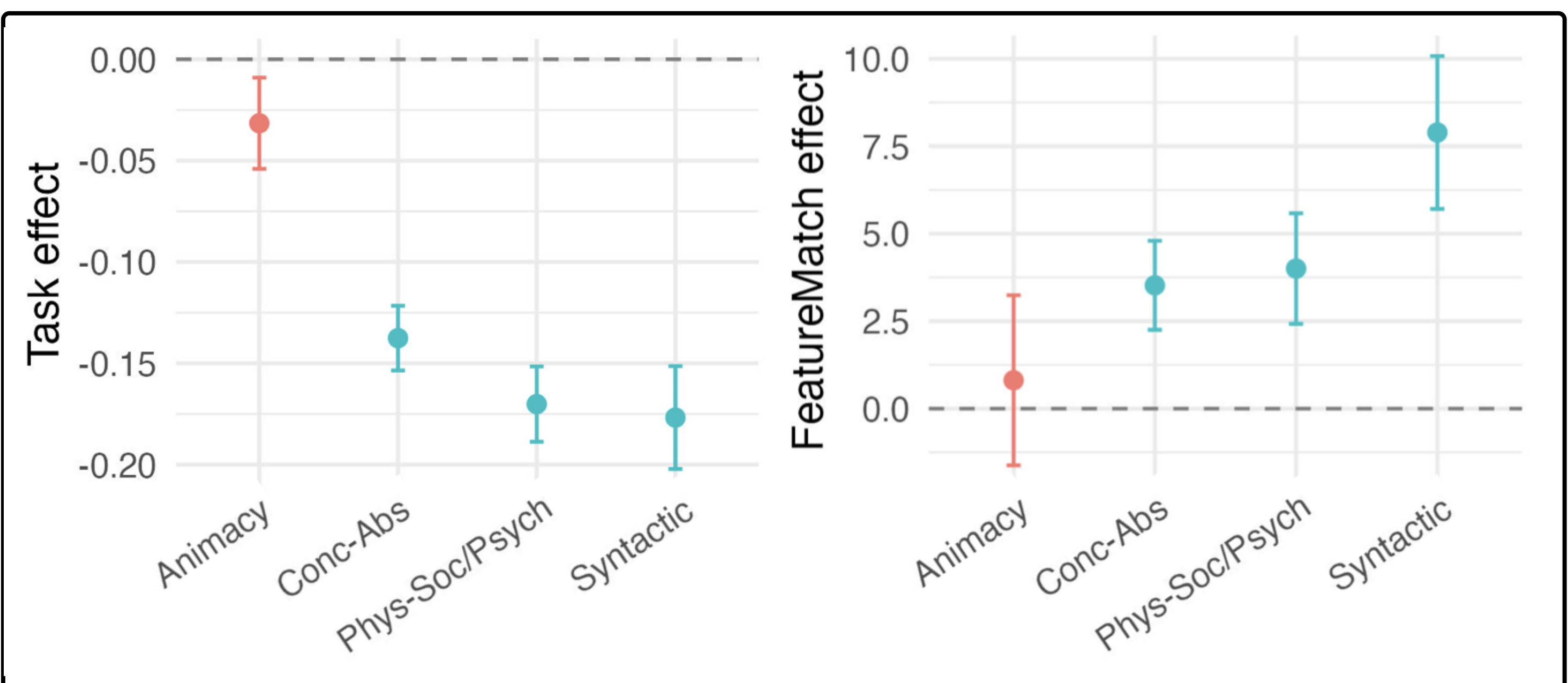


Similarity Judgment Results

ValenceMatch pairs more similar than **TruthCondMatch**

$\text{lmer}(\text{Rating} \sim \text{FeatureMatch} + (1 + \text{FeatureMatch} | \text{ID}) + (1 | \text{Word}))$

Animacy Strongest Competitor to Valence



VALENCE IS HIGHLY PROMINENT (H3)

- Participants categorized valence faster than most truth-conditional features and relied on valence more in similarity judgments.
- Pattern consistent across multiple domains: valence is accessed earlier and weighted more strongly.
- Animacy strongest non-valence competitor: not all truth-conditional features equally prominent.
- Results support a prominence-based view in which affective info is foundational to lexical meaning.

Thank you!

Research funded by Sorbonne Université Projet Emergence "REV" OP24-126 and ERC Advanced Grant "Valence Asymmetries", n° 101142133. Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

References & PDF

